

Geometric & Topological Methods in Atmospheric Dynamics

Reference Card — UAH Graduate Course

I. Differential Geometry

Objects on a manifold M

- Manifold M : smooth space locally $\cong \mathbb{R}^n$
- Tangent space $T_p M$: velocity vectors at p
- Cotangent space $T_p^* M$: dual; holds 1-forms

Graded algebra of k -forms Ω^k

- 0-form: scalar (p, T, θ)
- 1-form v : velocity / circulation d.o.f.
- 2-form $\omega = dv$: vorticity / flux d.o.f.

Key operators

Op. Action	Property
d	$\Omega^k \rightarrow \Omega^{k+1}$
$\wedge$	$\Omega^p \times \Omega^q \rightarrow \Omega^{p+q}$
$\star$	$\Omega^k \rightarrow \Omega^{n-k}$
ι_u	$\Omega^k \rightarrow \Omega^{k-1}$
$\mathcal{L}_u$	$\Omega^k \rightarrow \Omega^k$

Cartan's magic formula

$$\mathcal{L}_u \alpha = d(\iota_u \alpha) + \iota_u(d\alpha)$$

Stokes' theorem

$$\int_{\partial M} \omega = \int_M d\omega$$

II. Geometric Fluid Dynamics

Vector-invariant incompressible Euler

$$\partial_t v + \iota_u \omega + dB = 0, \quad \omega = dv$$

$$B = \frac{1}{2}|u|^2 + p + \Phi, \quad d(\star v) = 0$$

$$\partial_t \omega + \mathcal{L}_u \omega = 0$$

Conserved invariants

Invariant	Expression
Energy	$E = \frac{1}{2} \int_{\Omega} v \wedge \star v$
Kelvin circ.	$\frac{d}{dt} \oint_{C(t)} v = 0$
Helicity (3D)	$H = \int_{\Omega} v \wedge dv$
Pot. vorticity	$q = (\omega + f)/\rho$
Enstrophy (2D)	$Z = \frac{1}{2} \int \omega^2 dA$

III. Discrete Exterior Calculus

Philosophy: separate topology (combinatorial) from metric (geometric).

- **Primal complex:** Delaunay triangles; k -cochains
- **Dual complex:** Voronoi polygons; $(n-k)$ -cochains
- **Coboundary op.** D : discrete d ; $D^2 = 0$ exactly
- **Discrete Hodge** M_k : diagonal mass matrix; sole error source
- **Whitney forms:** cochain-to-form interpolation

Discrete energy identity

$$\langle v, I_v(\tilde{D}_1 v) \rangle_1 = 0$$

Unconditional stability; no artificial dissipation.

Applications: MPAS, ICON, mimetic FD, compatible FE.

IV. Dynamical Systems & Bifurcations

Local bifurcations

Type	Normal form	Example
Saddle-node	$\dot{x} = \mu - x^2$	tipping pt.
Pitchfork	$\dot{x} = \mu x - x^3$	sym. breaking
Transcritical	$\dot{x} = \mu x - x^2$	threshold
Hopf	$\dot{r} = \mu r - r^3$	oscillation

Global: homoclinic, heteroclinic, period-doubling, torus, crisis.

Catastrophe theory (Thom)

Catastrophe	Codim.	Relevance
Fold	1	sudden onset
Cusp	2	BL hysteresis
Swallowtail	3	multi-param. tip
Butterfly	4	—

V. Lagrangian Coherent Structures

Flow map: $F_{t_0}^t(x_0)$ — trajectory from x_0 at t_0 .

Cauchy–Green tensor

$$C = (DF_{t_0}^t)^\top DF_{t_0}^t$$

FTLE

$$\sigma_{t_0}^t(x_0) = \frac{1}{|t - t_0|} \ln \sqrt{\lambda_{\max}(C)}$$

LCS types

Type	FTLE	Geometry
Repelling	fwd ridges	unstable mflds
Attracting	bwd ridges	stable mflds
Elliptic	closed lines	vortex cores
Parabolic	jet surfaces	shear barriers

Applications: atmospheric rivers, hurricane eyewalls, jet streams, polar vortex edge, aerosol transport.

VI. Topological Data Analysis

Pipeline

$$\text{cloud} \xrightarrow{\text{VR}} \text{filtration} \xrightarrow{H_k} \text{barcodes}$$

- β_0 : connected components
 - β_1 : loops / vortex rings
 - β_2 : enclosed voids
 - Persistence diagram: birth–death pairs
 - Barcode: feature lifetime
 - Mapper: topological graph
- Applications:** convective organization, cloud morphology, turbulence intermittency, regime detection, radar analysis.

VII. Atmospheric Boundary Layer Geometry

Concept	Geometric meaning
M-O similarity	solution manifold
Stability regimes	folded equil. surface
Cusp bifurcation	BL hysteresis
Ri number	stability manifold coord.
Slow manifold	near-critical dynamics
Wave–turb. inter-act.	resonant tori

Key dimensionless numbers

$$Ri = \frac{(g/\theta_v) \partial_z \theta_v}{(\partial_z u)^2 + (\partial_z v)^2}$$

$$Ro = \frac{U}{fL} \quad Fr = \frac{U}{NH}$$